

ILSA B. AHMED

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SUMMARY

Computer Science and Linguistics graduate who builds backend systems, APIs, and data pipelines from first principles and validates them with data. Experience across transactional backend services, high-throughput data infrastructure, and research pipelines evaluated on quantitative metrics. Comfortable owning a project from design through deployment and collaborating across teams.

EXPERIENCE

Lawly — Software Engineer Intern

Jun 2026 – Present

- Contribute across the full stack of a production Next.js 15 / React 19 / TypeScript application with a Fastify + Node.js backend and PostgreSQL (NeonDB) via Drizzle ORM.
- Write test cases and fix bugs tracked on a Kanban board (Plaky), verifying fixes with engineers before release.
- Work with WorkOS authentication (OAuth + JWT sessions), Stripe Checkout/Connect, Supabase real-time messaging, AWS S3, Twilio, and Brevo; deployed on Vercel.

Treevah — Software Engineer Intern

May 2025 – Aug 2025

- Resolved bugs through structured testing of a Node.js / React web application, improving stability ahead of releases.
- Integrated OAuth sign-in with Microsoft accounts using the Microsoft Authentication Library (MSAL).
- Built a multi-folder view letting users navigate nested folders synced from OneDrive via the Microsoft Graph API.

Health Care Engineering Systems Center — Data Annotator

Aug 2025 – Present

- Label eye-gaze patterns in gaze-tracking data from children with developmental disorders to support research and model training.

PROJECTS

Brokerage Platform — Simulated Stock Trading Web App | C#, .NET, ASP.NET MVC

2026

- Built a .NET MVC backend with a ledger service that records and persists every account transaction.
- Developed a background service that ticks security prices in real time to simulate live market data.
- Implemented a background matching engine that processes and executes buy and sell orders.

Illini Solar Car — Software Engineer, Telemetry Team (Volunteer)

Aug 2025 – Dec 2025

- Built a Python asyncio pipeline ingesting 50,000+ CAN Bus sensor data points per minute into QuestDB; designed indexing and partitioning strategies to cut query latency.
- Containerized services with Docker; built GitHub Actions CI/CD for automated testing, linting, and image builds, reducing manual deployment steps.
- Designed a FastAPI REST layer and React dashboard for real-time anomaly detection on the telemetry data.

AutoWiki — Multi-Agent Article Generator | Python, LangGraph, LangChain, GPT-4

2024 – 2026

- Built a command-line tool that generates well-sourced, Wikipedia-style articles from a single topic, implementing Stanford's STORM algorithm from the paper without using the original code.
- Designed a LangGraph multi-agent pipeline where perspective agents (e.g., biomedical engineer, ethicist) research scraped sources and draft a multi-section article, addressing the weak middle sections of LLM long-form output.
- Separated retrieval, orchestration, and evaluation layers; measured quality with ROUGE/BLEU, and structured prompt engineering cut hallucination rate by 52% vs. baseline.

Forward Data Lab, UIUC — Data Engineering (Volunteer Research)

Jan 2023 – Apr 2023

- Built document-ranking pipelines with TF-IDF and vector similarity (scikit-learn, spaCy) under faculty supervision.
- Exposed model outputs via Flask REST APIs backed by PostgreSQL and MongoDB; evaluated with precision, recall, F1, and cosine similarity.

TECHNICAL SKILLS

Languages: Python, C#, JavaScript, TypeScript, Java, SQL

Backend & Frontend: ASP.NET MVC, FastAPI, Flask, Fastify, Node.js, React, Next.js (App Router), Tailwind CSS, REST API design

Data & Quantitative: PostgreSQL, MongoDB, QuestDB (time-series), Pandas, Polars, NumPy, scikit-learn, spaCy, PyTorch, precision/recall/F1, cosine similarity, ROUGE/BLEU

AI/LLM: LangChain, LangGraph, RAG systems, vector embeddings, prompt engineering, OpenAI API, AI coding tools (Cursor, Claude Code)

Infra & Tools: Docker, GitHub Actions (CI/CD), AWS EC2/S3, Azure (Blob Storage, App Services), Microsoft Graph API, Vercel, Stripe, WorkOS, MSAL, Supabase, Twilio, Git, Postman, Figma, Notion

EDUCATION

University of Illinois Urbana-Champaign — B.S. Computer Science and Linguistics

Certifications: CompTIA Security+ (2026) | Aspiritech Academy — Cybersecurity Trainee, Security+ & CMMC training (Jun 2026)